COMPUTER ORGANIZATION AND ARCHITECTURE ASSIGNMENT – 3

Name: Rushi Daulatkar **Roll No.: 59**

UID: 23016060 **CSBS (2ND Year)** **Date: 30-Sep-24**

Case Study 8085 and 8056 Microprocessors

Introduction Microprocessors are the core of modern computing systems, playing a key role in the processing of data and control of systems. Two significant microprocessors that shaped early computing are the **Intel 8085** and **Intel 8056**. In this case study, we will explore these microprocessors in detail, discussing their architecture, instruction sets, functionalities, and applications, with a focus on how they contributed to the evolution of computing systems.

1. Intel 8085 Microprocessor

1.1 Overview

The Intel 8085 is an 8bit microprocessor introduced in **1976** by Intel. It is a successor to the 8080 microprocessor and is primarily used in embedded systems. The 8085 gets its name from the 5V power supply that it requires.

1.2 Key Features

- **8bit data bus** Can process 8bit data at a time.
- **16bit address bus** Can address up to 64 KB of memory.
- **74 instructions** It supports 74 distinct instructions, covering data transfer, arithmetic, logic, and control operations.
- **5V power supply** Operates on a single +5V power supply.
- **Multiplexed Address/Data Bus** The lower 8 bits of the address bus are multiplexed with the data bus.
- **Clock Speed** Operates at a clock speed of 3 MHz

1.3 Architecture

The architecture of the 8085 consists of various components that enable its functionalities

- **Accumulator (A)** A special purpose register used for arithmetic and logic operations.
- **Registers (B, C, D, E, H, L)** Six general purpose registers that can store 8bit data. They can be paired (BC, DE, HL) for 16bit operations.
- **Program Counter (PC)** A 16bit register used to hold the address of the next instruction to be executed.

- **Stack Pointer (SP)** A 16bit register those points to the top of the stack, useful for subroutines and interrupts.
- **Flag Register** Contains five flags (Sign, Zero, Auxiliary Carry, Parity, and Carry) that indicate the result of operations.
- **Instruction Decoder** Interprets instructions and signals other components to execute them.
- **Timing and Control Unit** Generates timing signals for the operation of the processor.

1.4 Instruction Set

The 8085 has a relatively small and simple instruction set. It includes the following groups

- **Data Transfer Instructions** MOV, MVI, LDA, STA, etc.
- **Arithmetic Instructions** ADD, SUB, INR, DCR, etc.
- **Logical Instructions** AND, OR, XRA, CMP, etc.
- **Control Instructions** HLT, NOP, DI, EI, etc.
- **Branching Instructions** JMP, CALL, RET, etc.

1.5 Interrupt System

The 8085 microprocessor has five interrupt signals

- **TRAP** A non-maskable interrupt (highest priority).
- **RST7.5, RST6.5, RST5.5** Maskable interrupts with different priorities.
- **INTR** A general-purpose interrupt.

1.6 Applications

The Intel 8085 has been used in

- **Embedded Systems** Control units in industrial systems.
- **Communication Systems** Basic telecommunications.
- **Microcomputers** Early personal computers.

---

2. Intel 8086 Microprocessor

2.1 Overview

The Intel 8086 is a 16bit microprocessor introduced in 1978, which serves as the foundation for the x86 architecture. It is known for its powerful processing capabilities and versatility, making it suitable for a wide range of applications from personal computing to embedded systems.

2.2 Key Features

- 16bit CPU: Capable of handling 16bit operations and data.
- Memory Addressing: Supports up to 1 MB of memory through segmented memory architecture.
- I/O Ports: Up to 256 I/O ports for device communication.
- Clock Frequency: Typically operates at clock speeds ranging from 5 MHz to 10 MHz.
- Instruction Queue: Supports a 6byte instruction queue for improved performance.
- Interrupt Handling: Provides support for up to 256 interrupt vectors.
- Multiprocessing Support: Can be configured to support multiple processors in a system.

2.3 Architecture

The 8086 microprocessor features a segmented architecture, which includes:

Registers:

- Accumulator (AX): Used for arithmetic operations.
- Base Register (BX): Pointer for data storage.
- Counter Register (CX): Used for loop control and string operations.
- Data Register (DX): Used in multiplication and division operations.

Segment Registers:

- CS (Code Segment): Points to the segment containing the program code.
- DS (Data Segment): Points to the segment for data storage.
- SS (Stack Segment): Points to the segment for stack operations.
- ES (Extra Segment): Used for additional data storage.
- Instruction Pointer (IP): Points to the next instruction to be executed.
- Flags Register (EF): Contains status flags (Zero, Sign, Carry, Overflow, etc.).

2.4 Instruction Set

The 8086-instruction set is extensive and designed for versatile tasks:

- Data Transfer Instructions: MOV, PUSH, POP, etc.
- Arithmetic Instructions: ADD, SUB, MUL, DIV, INC, DEC, etc.
- Logical Instructions: AND, OR, XOR, NOT, etc.
- Control Transfer Instructions: JMP, CALL, RET, etc.
- String Instructions: MOVS, CMPS, SCAS, etc.

2.5 Interrupt System

The 8086 features a robust interrupt system:

- Hardware Interrupts: Responds to hardware events through interrupt requests (IRQ).
- Software Interrupts: Triggered by specific software instructions (INT).
- Trap and Fault Interrupts: Used for exception handling and debugging.

2.6 Applications

The 8086 microprocessors have been widely used in:

- Personal Computers: Forms the basis for many early PC architectures.
- Embedded Systems: Utilized in various consumer electronics.
- Industrial Automation: Controls machinery and processes in factories.
- Telecommunications: Supports communication protocols and systems.

---

3. Comparison between 8085 and 8086

Feature	Intel 8085	Intel 8086
Type	Microprocessor	Microprocessor
Architecture	8-bit	16-bit
Clock Speed	Up to 3 MHz	5 MHz to 10 MHz
Memory	16-bit address bus (up to 64 KB of external memory)	20-bit address bus (up to 1 MB of memory)
I/O Ports	8-bit I/O ports (64 I/O ports)	16-bit I/O ports (up to 256 I/O ports)
Timers/Counters	No built-in timers	No built-in timers
Interrupts	5 interrupt sources	256 interrupt sources
Applications	General-purpose computing	Personal computers, embedded systems, and complex applications

---

4. Conclusion

The Intel 8085 and 8086 microprocessors were significant milestones in the evolution of computing technology. The 8085, introduced in the mid-1970s, laid the groundwork for basic microprocessor architecture and general-purpose computing. In contrast, the 8086, launched in the late 1970s, marked a shift towards more advanced computing capabilities with its 16-bit architecture and support for larger memory addressing. Understanding their architectures, instruction sets, and applications provides valuable insights into the development of modern computing systems and the transition from simple microprocessors to more sophisticated processing units that power today's computers.

History of the 8085 and 8086 Microprocessors

1. Intel 8085 Microprocessor

1.1 Background and Development

The Intel 8085 microprocessor was developed in the mid1970s as part of Intel's initiative to create efficient microprocessors for data processing in embedded systems and general computing. It is a direct descendant of the Intel 8080, one of the first general-purpose microprocessors launched in 1974. The 8080 had established Intel's position in the microprocessor market, being a popular choice for early personal computers and embedded applications.

In 1976, Intel aimed to enhance the 8080 with a more integrated design and additional features to cater to the growing demands of embedded systems. The 8085 was introduced as a more cost-effective and power efficient alternative to its predecessor, featuring a simplified architecture. Its name reflects its compatibility with a 5V power supply, reducing the complexity of power requirements for microprocessor-based systems.

1.2 Milestones

1976: The 8085 was launched as an 8bit microprocessor with a simpler design and lower power consumption than the 8080.

The 8085 integrated the clock generator and system controller into a single chip, significantly reducing the number of external components required for operation.

It gained traction in early embedded systems, industrial control units, and telecommunications equipment.

While not as influential in personal computing as the 8080, the 8085 found its niche in educational systems and training kits for teaching microprocessor design and programming.

1.3 Influence on Later Technologies

The 8085's success in embedded systems paved the way for more advanced microprocessors like the Intel 8086, which was launched in 1978 and marked Intel's entry into the 16bit microprocessor market. The x86 architecture introduced by the 8086 would dominate the personal computer market for decades, building upon the foundation laid by the 8085.

2. Intel 8086 Microprocessor

2.1 Background and Development

The Intel 8086 microprocessor was developed as a response to the demand for more powerful processing capabilities in personal computing and complex applications. Launched in 1978, the 8086 was Intel's first 16bit microprocessor, designed to extend the capabilities of the earlier 8bit processors. Its architecture supported a 20bit address bus, allowing it to access up to 1 MB of memory, a significant leap from its predecessors.

The development of the 8086 was influenced by the growing market for personal computers and the need for a more advanced instruction set to support new applications. Intel aimed to create a microprocessor that could handle more complex tasks and support the emerging software ecosystem, leading to the design of the x86 architecture.

2.2 Milestones

1978: The 8086 was launched, featuring a 16bit architecture and a segmented memory model, which allowed for better organization and management of memory.

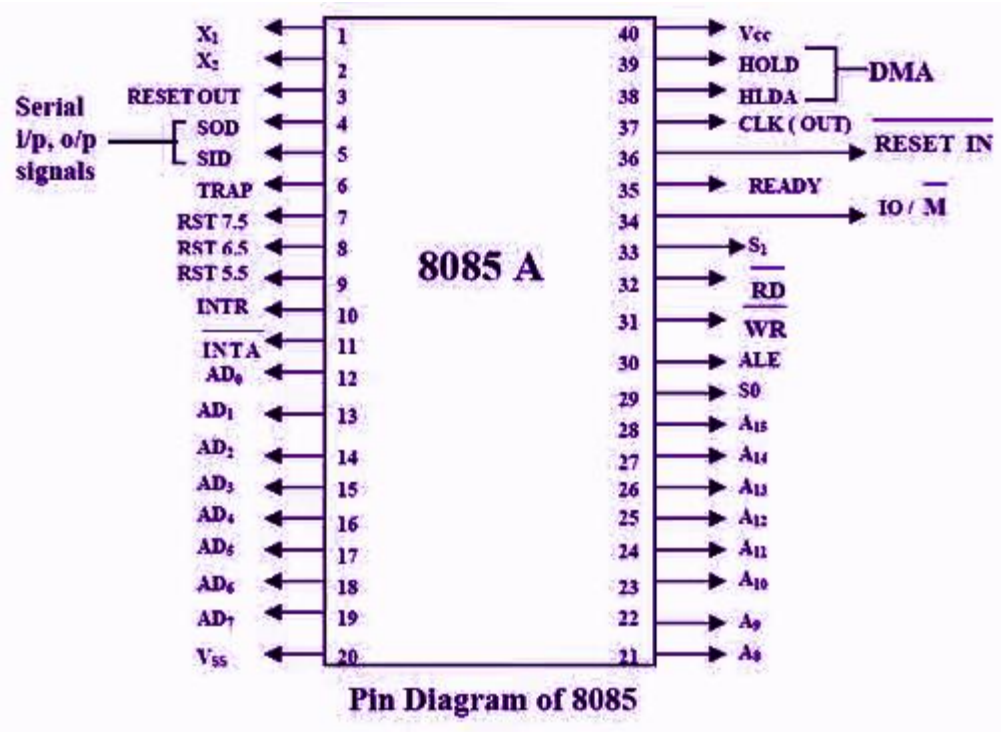
The introduction of the 8086 set the stage for the x86 architecture, which became the basis for a vast range of personal computers and compatible systems.

The 8086 gained popularity for its performance in business applications, gaming, and various software development environments, making it a key player in the early personal computing revolution.

2.3 Longevity and Legacy

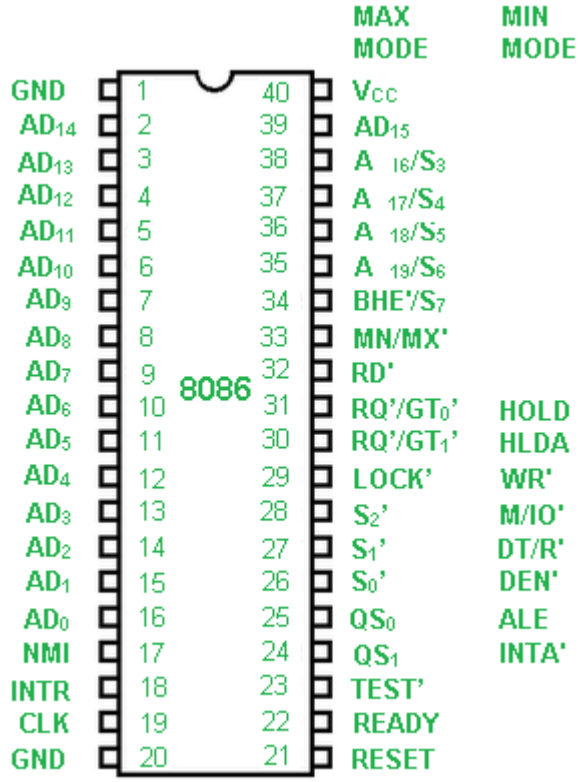
The Intel 8086 had a profound impact on the computing landscape, leading to the widespread adoption of the x86 architecture. Its design principles and instruction set formed the basis for future generations of processors, ensuring compatibility with a wide range of software applications. The 8086 remains relevant today, as the x86 architecture continues to be the foundation for modern personal computers, servers, and various computing devices.

Diagram of 8085:



Pin Diagram of 8085

Diagram of 8086:



Description of the 40 s of the Intel 8085 microprocessor:

1. X1: Input for external clock generator. Connected to a crystal or clock source to generate the clock signal.

2. X2: Output for the external clock. Works with 1 to generate the clock for the microprocessor.

3. RESET OUT: Indicates that the microprocessor is being reset. It can be used to reset external devices.

4. SOD (Serial Output Data): Serial output data line for transmitting data serially.

5. SID (Serial Input Data): Serial input data line for receiving data serially.

6. TRAP: A non-maskable interrupt with the highest priority. Cannot be disabled.

7. RST 7.5: Maskable, vectored interrupt with the highest priority among RST interrupts.

8. RST 6.5: Maskable, vectored interrupt with medium priority among RST interrupts.

9. RST 5.5: Maskable, vectored interrupt with the lowest priority among RST interrupts.

10. INTR (Interrupt Request): General-purpose, maskable interrupt request input.

11. INTA (Interrupt Acknowledge): Output signal that acknowledges the interrupt request from INTR.

12-19. AD0-AD7 (Multiplexed Address/Data Bus): These 8 s are multiplexed to carry the lower 8 bits of the address during the first part of the machine cycle, and then data during the second part.

20. GND: Ground reference for the microprocessor.

21-28. (Higher order Address Bus): These 8 s carry the most significant 8 bits of the address bus during memory operations.

29. S0: Status signal used to indicate the current machine cycle.

30. ALE (Address Latch Enable): Indicates that the lower 8 bits on AD0AD7 are address bits. Used to latch the address into an external latch.

31. IO/M (Input/Output Memory): Distinguishes whether the operation is related to memory (low) or I/O (high).

32. RD (Read): Active low signal indicating that the processor is performing a read operation from memory or an I/O device.

33. WR (Write): Active low signal indicating that the processor is performing a write operation to memory or an I/O device.

34. S1: Status signal used to indicate the current machine cycle (works with S0).

35. READY: Input signal indicating if the external device is ready for communication. The processor waits if the device is not ready.

36. RESET IN: Active high signal used to reset the microprocessor. Clears the program counter and resets the processor.

37. CLK OUT: Provides the clock output for synchronization with external devices.

38. HLDA (Hold Acknowledge): Hold Acknowledge signal. The processor acknowledges that it has received a HOLD request.

39. HOLD: Request signal for DMA (Direct Memory Access). When high, indicates that an external device is requesting control of the buses.

40. Vcc: +5V power supply.

Description of the 40 pins of the Intel 8086 microprocessor:

Left Side (120):

1. GND Ground.

2. AD14 Address/Data line 14 (multiplexed).

3. AD13 Address/Data line 13 (multiplexed).

4. AD12 Address/Data line 12 (multiplexed).

5. AD11 Address/Data line 11 (multiplexed).

6. AD10 Address/Data line 10 (multiplexed).

7. AD9 Address/Data line 9 (multiplexed).

8. AD8 Address/Data line 8 (multiplexed).

9. AD7 Address/Data line 7 (multiplexed).

10. AD6 Address/Data line 6 (multiplexed).

11. AD5 Address/Data line 5 (multiplexed).

12. AD4 Address/Data line 4 (multiplexed).

13. AD3 Address/Data line 3 (multiplexed).

14. AD2 Address/Data line 2 (multiplexed).

15. AD1 Address/Data line 1 (multiplexed).

16. AD0 Address/Data line 0 (multiplexed).

17. NMI non-maskable interrupt.

18. INTR Interrupt request.

19. CLK Clock input for timing and synchronization.

20. GND Ground.

Right Side (s 2140)

21. RESET Resets the processor.

22. READY Indicates whether external devices are ready for data transfer.

23. TEST Test input, used in conjunction with the WAIT instruction.

24. QS1 Queue status, used in maximum mode to indicate the status of the instruction queue.

25. QS0 Queue status, used in maximum mode to indicate the status of the instruction queue.

26. S0 Status signal (machine cycle status).

27. S1 Status signal (machine cycle status).

28. S2 Status signal (machine cycle status).

29. LOCK Bus lock signal, prevents other processors from accessing the bus.

30. RQ/GT1 Request/Grant line 1, used for bus arbitration in maximum mode.

31. RQ/GT0 Request/Grant line 0, used for bus arbitration in maximum mode.

32. **RD** is the Read control signal, used to read data from memory or input devices.

33. MN/MX Mode selection, high for minimum mode, low for maximum mode.

34. BHE/S7 Bus High Enable or status signal S7, indicates high byte access.

35,36,37,38. s 35 (A19/S6), 36 (A18/S5), 37 (A17/S4), and 38 (A16/S3) on the 8086 microprocessors provide the upper bits (1916) of the address bus and serve as status lines (S6S3). They are essential for address signalling and indicating the current operation status.

39. AD15is the Address/Data line 15, used for the 15th bit of the address and data transfer.

40. VCC +5V power supply

The instruction format of the 8085 microprocessors can be described as follows:

Instruction Format:

1. Opcode: Specifies the operation to be performed (1 to 3 bytes).

2. Operands: Data or address used by the operation (0 to 2 bytes).

Breakdown:

- Single Byte Instruction:
  - Example: `MOV A, B` (1 byte: opcode only)

- Two Byte Instruction:
  - Example: `MOV M, A` (1 byte: opcode + 1 byte: address)

- Three Byte Instruction:
  - Example: `LXI H, 2050H` (1 byte: opcode + 2 bytes: 16-bit address)

The instruction format of the 8086 microprocessors can be described as follows:

Instruction Format:

1. Opcode: Specifies the operation to be performed (1 to 2 bytes).
2. Operands: Data or address used by the operation (0 to 2 bytes).

Breakdown:

- Single Byte Instruction:
  - Example: `ADD AL, 5` (1 byte: opcode + 1 byte: immediate value)

- Two Byte Instruction:
  - Example: `MOV AX, BX` (1 byte: opcode + 1 byte: ModRM byte)

- Three Byte Instruction:
  - Example: `MOV CX, [1234H]` (1 byte: opcode + 1 byte: ModRM byte + 1 byte: displacement)

- Four Byte Instruction:
  - Example: `CALL 1234H` (1 byte: opcode + 2 bytes: offset + 1 byte: segment) .

Registers in the 8085 Microprocessor:

1. Accumulator (A): Used for arithmetic and logic operations.

2. Registers B, C, D, E, H, L: General-purpose registers for data storage and operations.

3. Program Counter (PC): Holds the address of the next instruction to be executed.

4. Stack Pointer (SP): Points to the top of the stack in memory.

5. Instruction Register (IR): Holds the opcode of the currently executing instruction.

6. Temporary Register: Used for temporary data storage during instruction execution.

Registers in the 8086 Microprocessor:

1. General-Purpose Registers:
   - AX (Accumulator): Used for arithmetic operations.
   - BX (Base Register): Used for addressing and as a pointer.
   - CX (Count Register): Used for loop control and counting operations.
   - DX (Data Register): Used for I/O operations and arithmetic operations.
2. Segment Registers:
   - CS (Code Segment): Points to the segment containing the currently executing code.
   - DS (Data Segment): Points to the segment containing data.
   - SS (Stack Segment): Points to the segment containing the stack.
   - ES (Extra Segment): Additional data segment used for operations.
3. Pointer and Index Registers:
   - SP (Stack Pointer): Points to the top of the stack.
   - BP (Base Pointer): Used for accessing parameters in the stack.
   - SI (Source Index): Used for string operations.
   - DI (Destination Index): Used for string operations.
4. Instruction Pointer (IP): Holds the offset address of the next instruction to be executed in the code segment.

5. Flags Register: Contains status flags that indicate the status of the processor and the results of operations.

Summary:

- 8085 has a simpler architecture with fewer registers, focusing on general-purpose operations.

- 8086 features a more complex architecture with multiple registers to support segmented memory addressing and more advanced operations.